MARYAM REHAN

Cyber Security Internship Report

Week 3 — Task 8

Cyber Kill Chain MITRE ATTACK Introduction

1. Objective

The objective of this task was to understand the Lockheed Martin Cyber Kill Chain framework and the MITRE ATT&CK framework. The task focused on identifying the seven stages of a cyberattack, understanding MITRE ATT&CK tactics and techniques, and mapping a Brute Force attack to the Cyber Kill Chain.

2. Cyber Kill Chain

Seven Stages

1. **Reconnaissance:** The attacker collects information about the target system or organization.
2. **Weaponization:** The attacker prepares malicious tools, payloads or resources for the attack.
3. **Delivery:** The attacker sends the prepared payload or initiates the attack against the target.
4. **Exploitation:** The attacker takes advantage of a vulnerability or weakness to gain access.
5. **Installation:** Malware or a backdoor may be installed to maintain access.
6. **Command and Control:** The attacker establishes communication with the compromised system.
7. **Actions on Objectives:** The attacker performs the final malicious activities, such as stealing data or disrupting services.

3. MITRE ATT&CK Framework

MITRE ATT&CK is a knowledge base that documents real-world adversary tactics and techniques. It helps security analysts understand attacker behaviour, investigate security alerts and improve threat detection.

Tactics describe the attacker's objectives, while techniques explain how those objectives are achieved.

Examples of ATT&CK tactics include Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Exfiltration and Impact.

4. Selected MITRE Technique — T1110 Brute Force

Technique ID: T1110

Technique Name: Brute Force

Tactic: Credential Access

Brute Force is a technique in which an attacker repeatedly attempts different passwords to gain unauthorized access to an account or system.

Its four sub-techniques are:

- T1110.001 — Password Guessing
- T1110.002 — Password Cracking
- T1110.003 — Password Spraying
- T1110.004 — Credential Stuffing

Detection Methods:

- Monitoring repeated failed login attempts.
- Identifying suspicious IP addresses.
- Investigating unusual authentication activity.
- Checking for failed login attempts followed by successful authentication.

Prevention Methods:

- Enabling Multi-Factor Authentication.
- Using strong and unique passwords.
- Applying account lockout policies.
- Monitoring system and authentication logs.

5. Cyber Kill Chain Mapping — Brute Force Attack

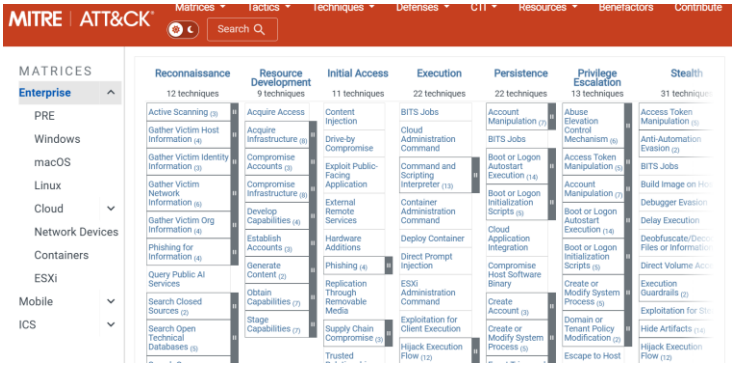
Scenario: An attacker attempts to gain unauthorized access to an organization's employee account through password guessing.

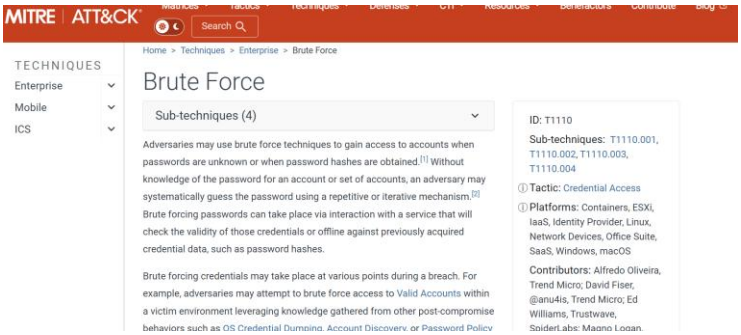
Cyber Kill Chain — 7 Stages and Activities

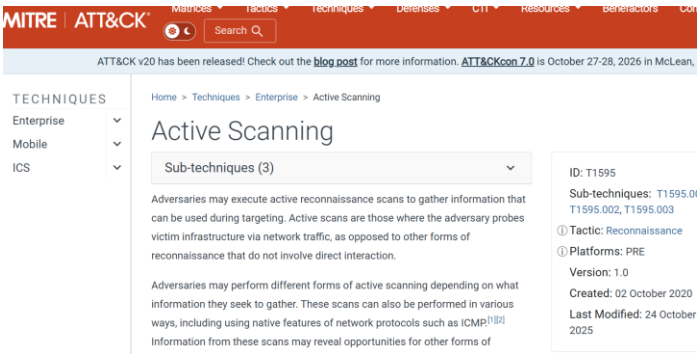
1. Reconnaissance
    - Collect information about the target.
    - Identify IP addresses, domains and employees.
    - Discover possible security weaknesses.
2. Weaponization
    - Prepare malicious files or payloads.
    - Select attack tools.
    - Create resources for the planned attack.
3. Delivery
    - Send malicious files through email.
    - Share harmful links.
    - Deliver payloads through websites or other channels.
4. Exploitation
    - Exploit vulnerabilities in the target system.
    - Take advantage of weak security.
    - Gain unauthorized access.
5. Installation
    - Install malware on the target system.
    - Create a backdoor.
    - Maintain unauthorized access.
6. Command and Control (C2)
    - Establish communication with the compromised system.
    - Remotely control the infected device.
    - Send commands to the compromised system.
7. Actions on Objectives
    - Steal confidential information.

- Disrupt system operations.
- Modify or destroy data.
- Achieve the attacker's final objective.

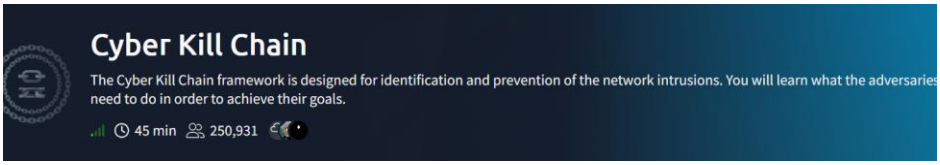
Evidence







Try Hack Me





T1

Learning Objectives

In this room, you will learn about each phase of the Cyber Kill Chain Framework, the advantages and disadvantages of the traditional Cyber Kill Chain.

Outcome
As a result, you will be ready to recognize different phases or stages of the attack carried out by an adversary and be able to break the "kill chain."

Answer the questions below

Read the above.

No answer needed — ✓ Correct Answer

T2

Answer the questions below

What is the name of the Intel Gathering Tool that is a web-based interface to the common tools and resources for open-source intelligence?

OSINT Framework — ✓ Correct Answer

What is the definition for the email gathering process during the stage of reconnaissance?

email harvesting — ✓ Correct Answer

T3

Answer the questions below

What is the term for automated scripts embedded in Microsoft Office documents that can be used to perform tasks or exploited by attackers for malicious purposes?

Macro ✓ Correct Answer

T4

Answer the questions below

What do you call an attack targeting a specific group by infecting their frequently visited website?

Watering hole attack ✓ Correct Answer

T5

Answer the questions below

What is the term for a cyber attack that exploits a software vulnerability that is unknown by software vendors?

Zero-day ✓ Correct Answer

T6

Answer the questions below

What technique is used to modify file time attributes to hide new or changes to existing files?

Timestomping ✓ Correct Answer

What malicious script can be planted by an attacker on the web server to maintain access to the compromised system and enables the web server to be accessed remotely?

Web shell ✓ Correct Answer

T7

Answer the questions below

What is the C2 communication where the victim makes regular DNS requests to a DNS server and domain which belong to an attacker.

DNS Tunneling ✓ Correct Answer

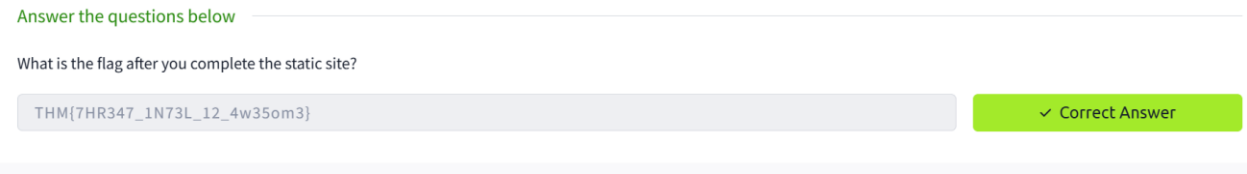
T8

Answer the questions below

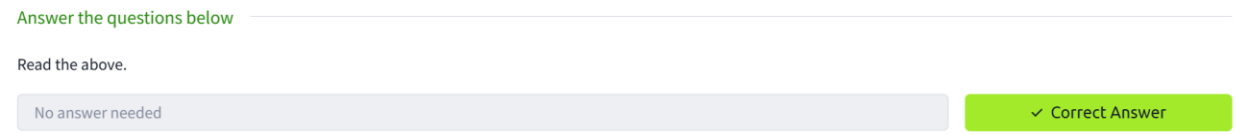
What technology is included in Microsoft Windows that can create backup copies or snapshots of files or volumes on the computer, even when they are in use?

Shadow Copy ✓ Correct Answer

T9



T10



Cyber Kill Chain Practical Learning Report

1. Introduction

The objective of this activity was to understand the Cyber Kill Chain framework and learn how cyberattacks progress through different stages.

2. Reconnaissance

Learned how attackers gather information about targets using OSINT tools, email harvesting, and active scanning.

3. Weaponization

Studied how attackers prepare malicious files, scripts, and payloads, including the misuse of Microsoft Office macros and PowerShell.

4. Delivery

Learned how attackers deliver malicious content through phishing emails, spearphishing attachments, and watering hole attacks.

5. Exploitation

Understood how attackers exploit software vulnerabilities, including zero-day vulnerabilities, to gain unauthorized access.

6. Installation

Learned how attackers maintain access through techniques such as web shells and timestomping.

7. Command & Control

Studied how attackers communicate with compromised systems using methods such as DNS tunneling and fallback channels.

8. Actions on Objectives (Exfiltration)

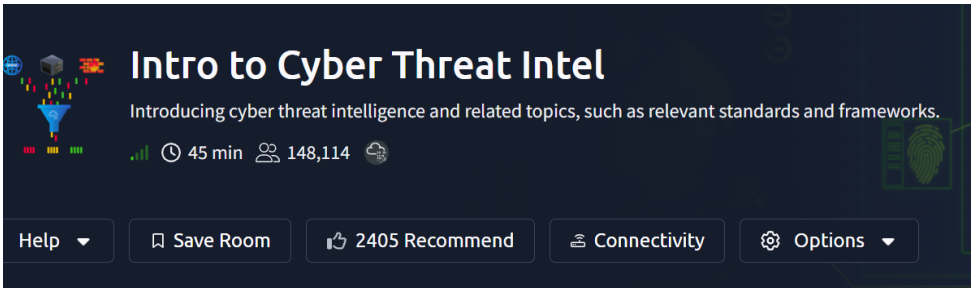
Learned how attackers collect sensitive information, create copies of data, and exfiltrate it from compromised systems.

9. Practice Analysis

Applied the Cyber Kill Chain framework to analyze the 2013 Target data breach scenario. Mapped six relevant techniques to their corresponding stages and completed the static-site exercise.

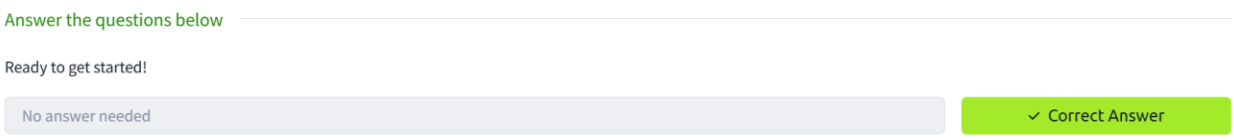
10. Conclusion

This practical improved my understanding of cyberattack stages, attacker techniques, and MITRE ATT&CK concepts. It also helped me develop basic skills in analyzing real-world cyber incidents using the Cyber Kill Chain framework.





T1



T2

Answer the questions below

What does CTI stand for?

Cyber Threat Intelligence ✓ Correct Answer

IP addresses, Hashes and other threat artefacts would be found under which Threat Intelligence classification?

Technical Intel ✓ Correct Answer

T3

Answer the questions below

At which phase of the CTI lifecycle is data converted into usable formats through sorting, organising, correlation and presentation?

Processing ✓ Correct Answer

During which phase do security analysts get the chance to define the questions to investigate incidents?

Direction ✓ Correct Answer

T4

Answer the questions below

What sharing models are supported by TAXII?

Collection and Channel ✓ Correct Answer

When an adversary has obtained access to a network and is extracting data, what phase of the kill chain are they on?

Actions on Objectives ✓ Correct Answer

T5

Answer the questions below

What was the source email address?

vipivillain@badbank.com ✓ Correct Answer

What was the name of the file downloaded?

flbpfuh.exe ✓ Correct Answer

After building the threat profile, what message do you receive?

THM{NOW_I_CAN_CTI} ✓ Correct Answer

Introduction to Cyber Threat Intelligence – Practical Report

Task 1: Introduction

Learned the importance of Cyber Threat Intelligence (CTI) in identifying threats, investigating alerts, and supporting SOC analysts in making informed security decisions.

Task 2: Cyber Threat Intelligence

Studied the difference between data, information, and intelligence. Learned about Indicators of Compromise (IOCs), Indicators of Attack (IOAs), and Tactics, Techniques, and Procedures (TTPs). Also explored intelligence sources and classifications, including technical, tactical, operational, and strategic intelligence.

Task 3: CTI Lifecycle

Learned the six phases of the CTI lifecycle:

- Direction
- Collection
- Processing
- Analysis
- Dissemination
- Feedback

Understood how raw threat data is collected, processed, analysed, and shared to support security decisions.

Task 4: CTI Standards & Frameworks

Explored MITRE ATT&CK, MITRE D3FEND, and the Cyber Kill Chain. Studied CVE, CVSS, NVD, STIX, and TAXII, including the TAXII sharing models: Collection and Channel.

Task 5: Practical Analysis

Analysed a simulated SIEM dashboard to investigate a suspicious email and related system activity.

Key Findings:

- Source Email: vipivillain@badbank.com
- Downloaded File: flbpfuh.exe
- Suspicious IP Address: 91.185.23.222
- Logged-in Account: Administrator
- Targeted User: John Doe

Practical Flag: THM{NOW_I_CAN_CTI}

Conclusion

This practical improved my understanding of cyber threat intelligence, its lifecycle, threat indicators, and security frameworks. It also provided experience in analysing SIEM logs and building a basic threat profile to support SOC investigations.